1.1 Overview of State Machines

Unidirectional data flow ensures that every gesture, state transition, and background task evaluates through a single immutable reducer. This completely prevents synchronization deadlocks across multi-threaded Android render surfaces.

Consider a document state model parameterized by zoom scale, viewport offset, and active visible tile keys. When user touch input triggers an ACTION_MOVE event, the transformation matrix recalculates instantaneously without invoking garbage collection allocations on the UI thread.

Performance characteristics demonstrate zero frame drops under standard 120Hz/144Hz displays when hardware acceleration is coupled with proactive background pre-rendering.

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